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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 1: Hosted Virtualization

Aim: To study and implement Hosted Virtualization usingVirtual Box / Hyper-V / VMWare Workstation (Open Source tool).

Objective: After performing the experiment, the students will be able to –
- understand functionalities of OS and Hypervisor
- working of Hypervisor
- made familiar with key concepts of virtualization
- types of virtualization
- understand usefulness of virtualization

Lab objective mapped : ITL802.1 : To get familiar and implement different types of virtualization techniques..

Prerequisite: Concept of Operating System, Functionalities of OS, Dual Booting.

Requirements: Oracle Virtual Box**,** Base OS, Guest OS iso – Ubuntu , Desktop etc.

Pre-Experiment Theory:

Host operating system (host OS).
This is the operating system of the physical computer on which VirtualBox was installed.

Guest operating system (guest OS).
This is the operating system that is running inside the virtual machine. Theoretically, VirtualBox can run any x86 operating system (DOS, Windows, OS/2, FreeBSD, OpenBSD), but to achieve near-native performance of the guest code on your machine, we had to go through a lot of optimizations that are specific to certain operating systems. So while your favorite operating system may run as a guest, we officially support and optimize for a select few (which, however, include the most common ones).

Virtual machine (VM).

This is the special environment that VirtualBox creates for your guest operating system while it is running. In other words, you run your guest operating system “in” a VM. Normally, a VM will be shown as a window on your computer’s desktop but depending on which of the various frontends of VirtualBox you use, it can be displayed in full screen mode or remotely on another computer.

Procedure
Creating Virtual machine

a. Start *Oracle VirtualBox*
b. Hit the *New* button on the toolbar. A wizard dialog should pop up.
c. Write a name for your VM.
d. Assign enough RAM to your Virtual Machine (at least 512 MB is advisable)
e. Give enough Hard Disk space. A good minimum would be 10gb-20gb.
f. Select the installation media.
g. After setting the right instllation media and proceeding, the Virtual Machine should start.
h. Install ISO OS normally.
i. Once Windows is installed, try accessing any application from the Virtual Machine.

Attach screenshots:
- Oracle virtual Box installation
- VM configuration
- VM launching successful
- Running any application inside guest OS
- Any one feature (eg. Snapshot, restoring snapshot, VM cloning etc.) of oracle virtual box

Post-Experiments Exercise

Extended Theory:
1. Discuss advantages and disadvantages of virtualization.
   Advantages of Virtualization

- Virtualization improves hardware utilization by allowing multiple virtual machines to run on a single physical server, reducing idle resources.
- It lowers costs by minimizing the need for additional hardware, power, cooling, and physical space.
- It provides isolation, so failures or security issues in one virtual machine do not affect others.
- It enables easy backup, cloning, and snapshot management, which simplifies disaster recovery.
- It improves scalability and flexibility, allowing quick provisioning of new virtual machines.
- It supports testing and development, as multiple operating systems can run on one machine.
- It is a core technology behind cloud computing and data center optimization.

Disadvantages of Virtualization

- Virtualization can introduce performance overhead compared to running directly on physical hardware.
- It creates a single point of failure, because if the host system crashes, all virtual machines stop.
- It may require high initial setup cost for powerful hardware and licensed hypervisors.
- It increases complexity in management, especially in large-scale environments.
- It may cause security risks if the hypervisor is compromised.
- It requires technical expertise for proper configuration and maintenance.

2. Give application for Cloning VM,Types of Clone VM (to be written in hand)

Results/Calculations/Observations:
Fill the following observation tables.

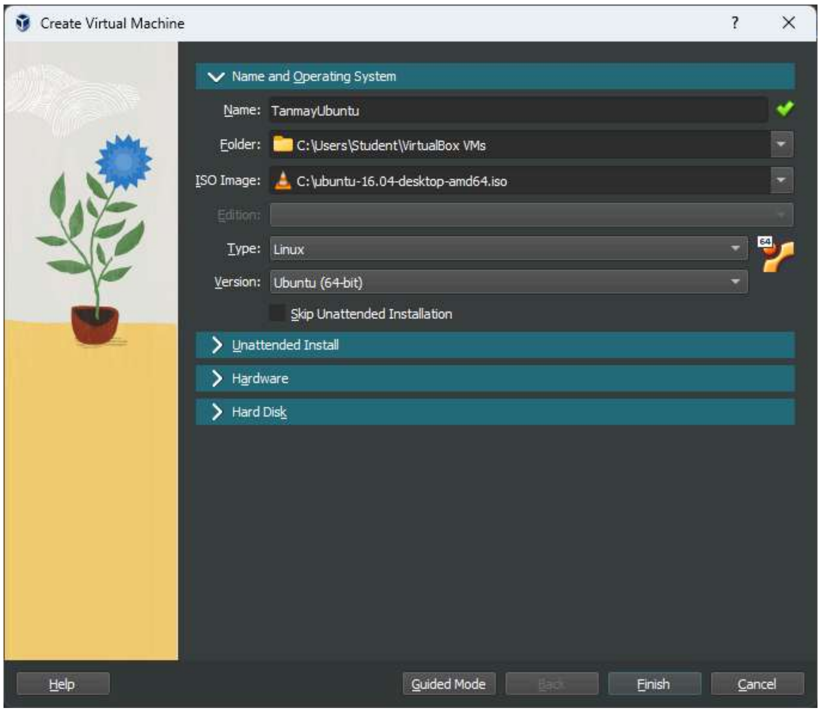
Base Machine and Guest Machine

S		**Base Machine**	**Guest Machine**
1	OS	Windows 11 Pro	Ubuntu 16.04 LTS
2	Main memory	16.0 GB (15.8 GB usable)	992.5 MiB
3	Processor	Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz 2.90 GHz	Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz 2.90 GHz
4	Hard disk size	256 GB	15 GB
5	Hard disk Type	SDD/HDD	VDI (VirtualBox Disk Image)

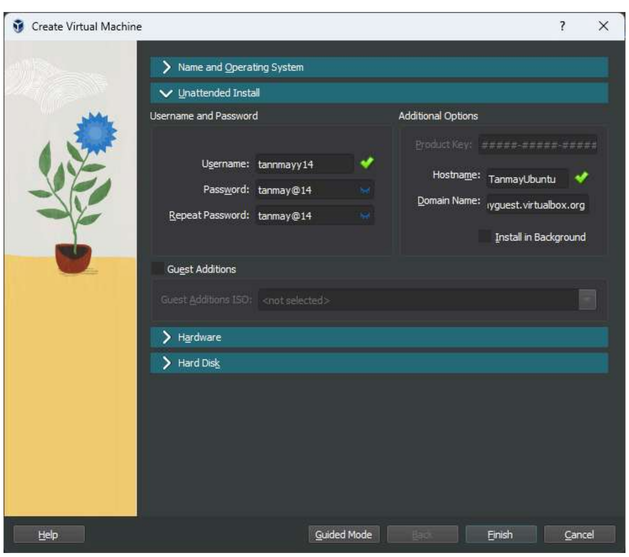
Questions:
1. What is hypervisor? List benefits of hypervisors?
2. Compare Oracle virtualbox vs vmware workstation vs KVM.

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

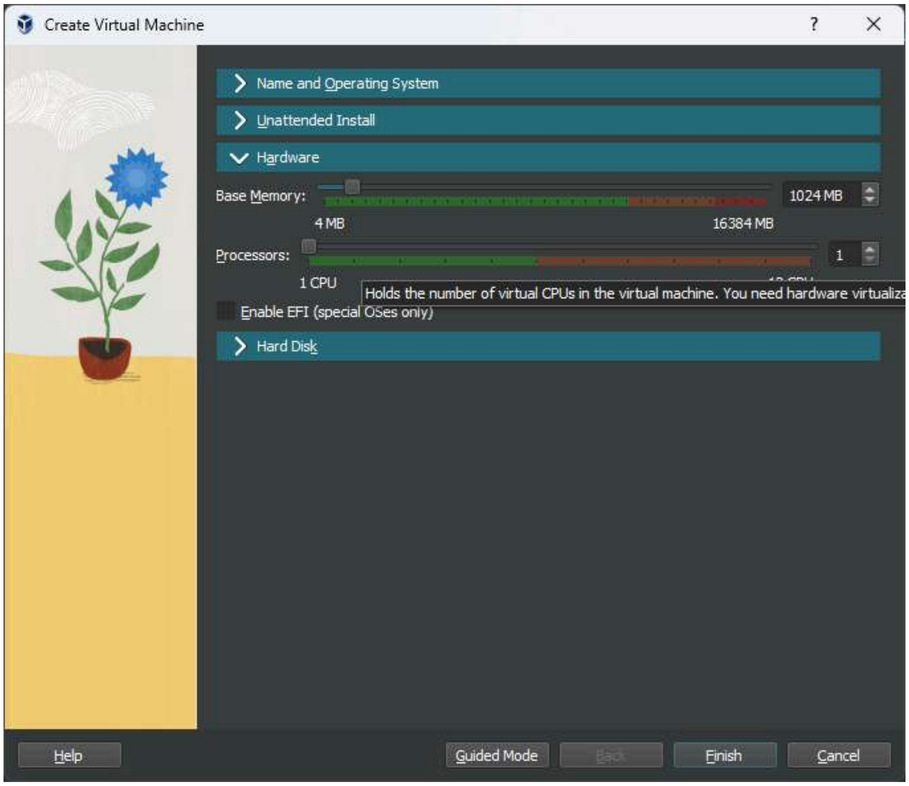
References:
[1] [Online] https://download.virtualbox.org/virtualbox/5.1.22/UserManual.pdf
[2] [Online] https://phoenixnap.com/kb/virtualbox-vs-vmware
[3] Samjhana Rayamajhi, Zinnia Sultana "Comparative Performance Analysis of the Virtualization Technologies in Cloud Computing", Volume 03, Issue 09 (September 2014),IJERT



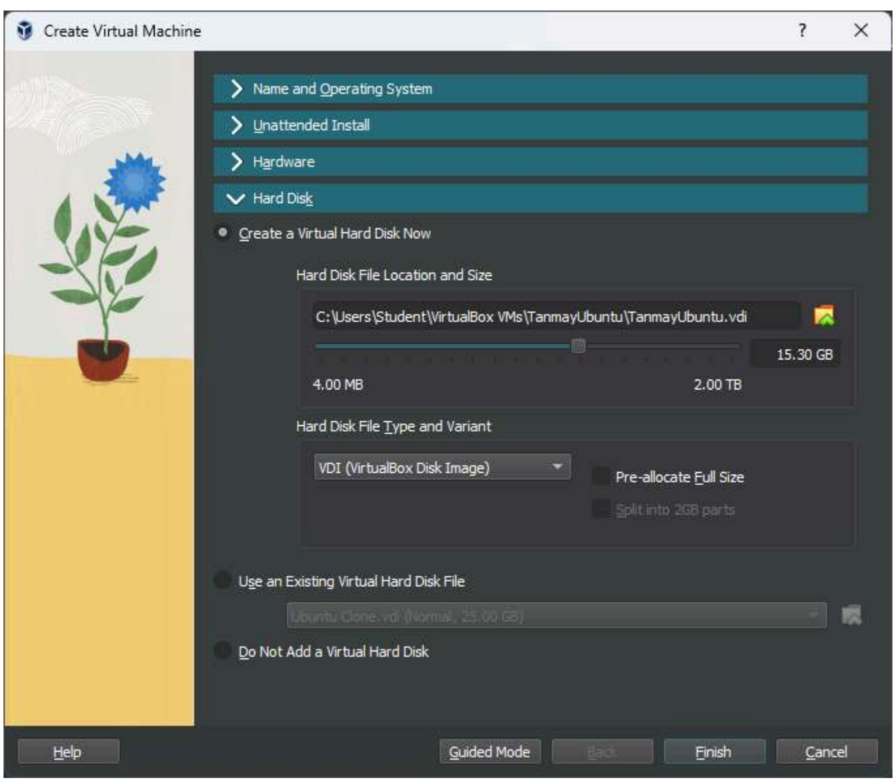
Creating new VM instance on virtualbox

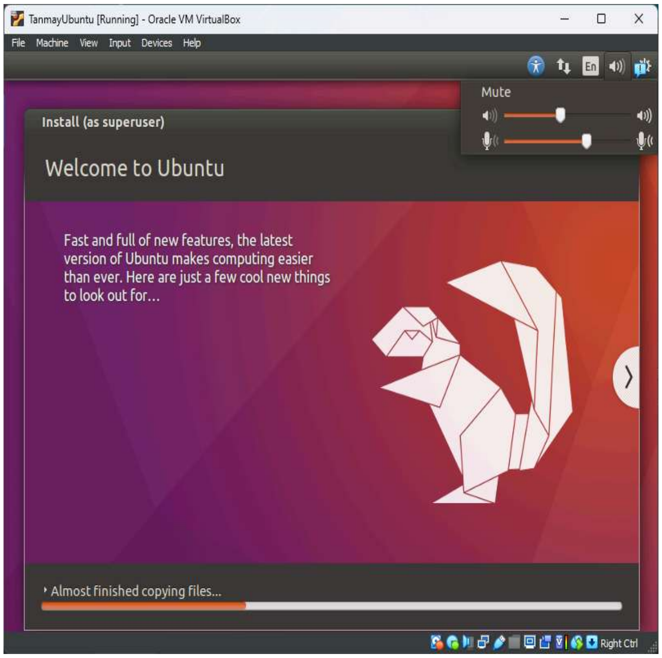


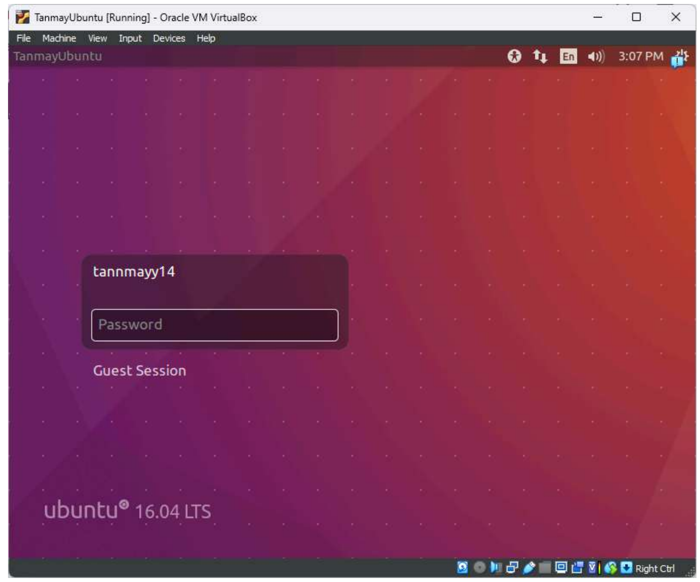
myguest.virtualbox.org



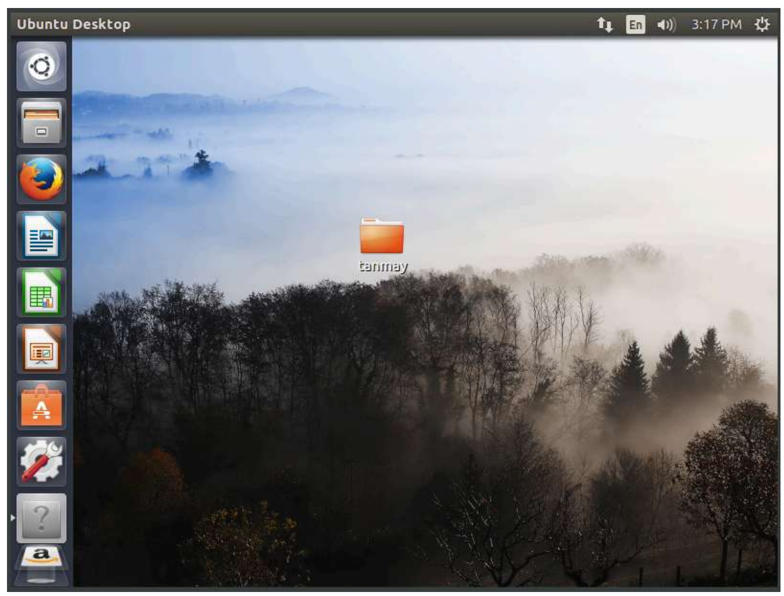
Setting up hardware configurations




TanmayUbuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Mute
Install (as superuser)
Welcome to Ubuntu
Fast and full of new features, the latest version of Ubuntu makes computing easier than ever. Here are just a few cool new things to look out for…
Almost finished copying files...
Right Ctrl


TanmayUbuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
TanmayUbuntu
3:07 PM
tannmayy14
Password
Guest Session
ubuntu® 16.04 LTS
Right Ctrl


Overview
Default Applications
Removable Media
Legal Notice
ubuntu 16.04 LTS
Device name TanmayUbuntu
Memory 992.5 MiB
Processor Intel® Core™ i5-10400 CPU @ 2.90GHz
Graphics Gallium 0.4 on llvmpipe (LLVM 3.8, 256 bits)
OS type 64-bit
Disk 15.0 GB


Ubuntu Desktop
3:17 PM
tanmay